

Curriculum Vitae, M. Haque

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Employment & Education

- **Current institute:** (since October 2006)
Max-Planck Institute for Physics of Complex Systems (MPI-PKS), Dresden, Germany.
Currently Staff Scientist (fixed-term) at MPI-PKS.
Former position title: *Distinguished PKS Postdoctoral Fellow*.
Held one of three such fellowships at MPI-PKS, until September 2009.
- Nov. 2003 to Sept. 2006: Postdoctoral researcher, Utrecht University, the Netherlands. Employed in cold-atom theory group led by H. T. C. Stoof.
- Ph.D., October 2003. Rutgers, State University of New Jersey.
Advisor: Andrei E. Ruckenstein.
Thesis title: *Interaction Effects in the Weakly Repulsive Bose Gas*.
- Graduate student and teaching assistant, 1997 to 2003.
Rutgers, State University of New Jersey; Department of Physics & Astronomy.
- Lecturer, 1996-97. Physics Department, Shahjalal University, Bangladesh.
- M.Sc., 1996; B.Sc., 1995. Shahjalal University, Bangladesh.

Track Record: Summary & Overview

- **Publications and citations (December 2011):** ResearcherID: C-4861-2011
Publications and preprints: 36. 7 papers in Phys. Rev. Lett.
Citations: 580⁺ on <http://adsabs.harvard.edu/>; 440⁺ on <http://www.researcherid.com/>.
- **Research Supervision:** Supervised or co-supervised postdocs, PhD students, undergraduate students. More than 10 publications from such supervisions. Details on p. 3.
- **Organization:** Organized two-week workshop in 2010; organizing one-week focus workshop scheduled 2012; preparing for a large 2013 workshop. Details on p. 2.
- **Research Area:** Condensed Matter Theory.
Major themes: (a) ultracold atoms; (b) non-equilibrium phenomena;
(c) entanglement in many-particle states, particularly topologically ordered states.

Research Interests

- **Area:** Condensed Matter Theory.
Collective phenomena in electronic, nanoscale and cold-atom systems.
- **Specific topics of published research:**
Non-equilibrium phenomena; collective quantum dynamics; quenches & ramps.
Cross-discipline application of concepts from *Quantum Information Theory* (*e.g.*, entanglement measures) to condensed matter issues.
Trapped atomic fermions and bosons — polarized fermion pairing; Feshbach resonances; BCS-BEC crossover; vortex dynamics in Bose condensates.
Topological order and unconventional states in many-particle systems.
Fractional quantum Hall states. Frustrated magnets. Spin chains.
Wigner crystallization. Pomeranchuk instabilities.
Laser-excited semiconductors; excitons. Nonlinear dynamics.

Organization activities

Workshop in June 2010

Title: Quantum information concepts for condensed matter problems (QICCMP)
Co-organizers: Ian Affleck (UBC, Canada); Ulrich Schollwöck (LMU, Munich).
Format: Two-week programme (June 14–25, 2010). 100⁺ participants.
Location & Funding: Max Planck Institute (MPI-PKS), Dresden, Germany.
Website: <http://www.pks.mpg.de/~qiccmp10/>

I conceived this event, assembled the organization team, and handled most details as the main organizer and contact person.

Workshop in November 2012

Title: Entanglement Spectra in Complex Quantum Wavefunctions (ESiCQW)
Co-organizers: B. Andrei Bernevig (Princeton); Andreas Läuchli (Innsbruck).
Format: One-week high-intensity workshop (November 12–16, 2012).
Location & Funding: Max Planck Institute (MPI-PKS), Dresden, Germany.

Other initiatives

Organizing various lectures, talks, journal clubs, locally, *e.g.*,

- a series of talks on “Cold atoms & BEC” in Dresden, 2008–2010.
- a journal club on cold-atom experiments, in Dresden, 2011.
- a lecture series on “1D & Exact Integrability” in Utrecht, spring 2006.
- the condensed matter theory seminar sequence at Utrecht, 2004–2006.

..... *etc.*

Research Supervision

Citations refer to publication list.

- **Publication Record of Supervision:** Supervised or co-supervised junior researchers for papers [8, 10, 11, 12, 14, 15, 16, 19, 20, 32, 33, 34].

- **PhD students and postdocs:**

Oleksandr Zozulya. Former PhD student of K. Schoutens at University of Amsterdam. I supervised his PhD research closely; all his PhD publications are coauthored with me [9, 10, 15, 19, 20].

Weibin Li. Former postdoc at MPI-PKS. Worked with me on vortex dynamics [16].

Vincenzo Alba. Postdoc at MPI-PKS (Oct. 2010 – present). Joint supervision with A. M. Läuchli. Research on entanglement in one-dimensional systems ([33]+ongoing).

Alexey Mikaberidze. PhD student at MPI-PKS till July 2011. Worked with me on an ecology-motivated calculation [14]. Based on this work, Mikaberidze is moving to a career in theoretical ecology after his PhD in atomic/laser physics.

Frank Zimmer. Postdoc at MPI-PKS (2009-2011). Worked with me on non-equilibrium dynamics (non-adiabatic ramps in trapped gases) [32, 34].

Ivana Vidanovic. PhD student in Belgrade. Working with me on non-equilibrium dynamics in spinor condensates.

Ricardo Pinto. Former PhD student at MPI-PKS. Worked with me on non-equilibrium dynamics in 1D lattice models (interaction-induced edge effects) [11].

- **Undergraduate Research Supervision (longer project):**

April 2011 – present: W. Tschischik, student from Technische Universität Chemnitz, pursuing one-year ‘Diplomarbeit’ research with me at MPI-PKS. Research on non-equilibrium dynamics in Bose-Hubbard ladders.

- **Undergraduate Research Supervision (short projects):**

Supervising summer research projects for visiting undergraduates.

Summer 2009: T. Venumadhav, visited Dresden from I.I.T. (Kanpur, India); 10 weeks. Research on non-equilibrium dynamics, led to publication; paper [8]. Venumadhav is now pursuing doctoral studies at Caltech.

February–July 2009: S. Piatecki, visited Dresden from ENS Paris; 5 months. Research on geometrically frustrated itinerant systems, later continued with additional collaborators ([2]+ongoing). Piatecki is now pursuing doctoral studies at ENS Paris.

Summer 2010: A. Balram, visited Dresden from ISERC (Pune, India); 3 months. Research on non-equilibrium dynamics in Bose-Hubbard rings. Balram is now pursuing doctoral studies at Penn State University (Pittsburgh).

Summer 2011: S. Chatterjee, visiting Dresden from I.I.T. (Kanpur, India); 10 weeks. Research on non-equilibrium dynamics; continued Balram’s work.

Grants & Funding

- **Workshop Funding** for 2-week event in June 2010:
 - ★ Funded by the conference program of the Max Planck Institute - PKS (Dresden).
 - ★ Accommodation for 100⁺ participants.
 - ★ Travel funding, total 15000 euros.
- **Grants** for collaboration visits, from several Research Networking Programmes of the European Science Foundation (ESF). Total 9 grants.
[Programme acronyms shown in brackets.]
 - ★ *from Dresden to London*: November 2011 [POLATOM].
 - ★ *from Dresden to Paris*: Oct. 2011 [QSpICE]; March 2011 [INTELBIOMAT]; Feb. 2010 [INTELBIOMAT]; Dec. 2009 [INSTANS].
 - ★ *Dresden to Amsterdam*: April 2009 [INSTANS] and November 2006 [INSTANS].
 - ★ *Dresden to Bristol*: October/November 2008 [HFM].
 - ★ *Utrecht to U.K. (Cambridge & Birmingham)*: May 2006 [INSTANS].

Funded visits were of durations between 6 and 15 days;
funding amounts were between 700 and 1400 euros.

Papers [9, 15, 17, 18, 19] resulted partly from these visits.

Teaching Experience

- **1997-2002**: Teaching Assistant, Physics & Astronomy Dept, Rutgers.
Teaching physics *recitations* (interactive problem sessions) and laboratories;
course administration of large courses (several hundred students).
Course administration involved coordination of testing and grading across 15-20 recitation sections taught by separate instructors, administration of midterm and final exams, organization and electronic implementation of grading schemes.
- **1996-1997**: Lecturer, Dept of Physics, Shahjalal University, Bangladesh.
Full-time teaching position, similar to teaching in a U.S. 4-year college.
Full responsibility for Electromagnetism class for beginning Physics majors. Partial responsibility for two advanced physics classes (Nuclear Physics, Atomic Physics).
Laboratory teaching.

Talks and Seminars

Invited presentations in Conferences & Workshops

- NORDITA conference (September 2010, Stockholm, Sweden):
Quantum Matter in Low Dimensions: Opportunities and Challenges.
Talk title: *Interaction induced hierarchy of edge-locking effects.*
- ESF Research Conference (June 2010, Obergugl, Austria):
Quantum Engineering of States and Devices: Theory and Experiments.
Talk title: *Edge-locking and Quantum Control in 1D lattice models.*
- ICTP workshop (May 2009, Trieste, Italy): *Research Frontiers in Ultracold Atoms.*
Talk title: *The Frontier of the Few: few-particle, few-vortex, and few-site surprises.*
- Dresden Workshop (March 2008): *Chaos and Collectivity in Many-Body Systems.*
Talk title: *A vortex dipole in a trapped 2D condensate.*
- Amsterdam summer workshop (July '07): *Low-D Quantum Condensed Matter 2007.*
Talk title: *Quantum information concepts for many-particle physics.*
- Mathematical Physics conference, September 2007, Tyler, Texas.
Talk title: *Entanglement measures used to probe topologically ordered states.*
- Dutch National Seminar (Landelijk Seminarium) on Statistical Physics (Nov. 2005).
Talk title: *Reaction front and pattern formation in an electronic system.*

Invited Seminars at various institutes and departments (month/year in parenthesis):

CEA Saclay, Paris (10/11); T.U. Dresden [Strunz group] (05/11); Lyon ENS (03/11);
 Cornell [Ithaca, N.Y.] (10/10); Boston University (10/10);
 Madrid Complutense (10/10); Toulouse (05/10); Oxford (03/10);
 Munich LMU (02/10); Barcelona [Casteldefels] ICFO (01/10);
 Paris Jussieu [LPTMC] (12/09); Grenoble ILL (11/09);
 Dresden MPI-PKS (09/09, 11/06); Bristol (11/08, 09/07); ENS Paris (02/08);
 M.I.T. (09/07); Florida Tallahassee (09/07); Rutgers (09/07, 12/05);
 Düsseldorf (03/07); Groningen (01/07); Utrecht (11/06, 12/03);
 Amsterdam (11/06); Cambridge [Cavendish] (01/06);
 Rutherford Lab [ISIS] (11/05); Karlsruhe (07/05); Frankfurt (07/05);
 ETH Zurich (12/04); Leiden (02/04).

Publication List

• Overview & highlights:

- Citations: 580⁺, in December 2011.
For selected papers, individual citation counts (December 2011) are noted below.
Citation data from <http://adsabs.harvard.edu/>
- Single-author publications: 4, 13, 24.
- Articles in higher-profile journals:
In *Physical Review Letters*: 1, 2, 5, 20, 21, 22, 26.
In *New Journal of Physics*: 6.
- Paper 9 is a review article.
- Articles 3 and 22 are coauthored with experimental groups.
- Articles highlighted with ▲ were coauthored solely with junior scientists or peers, supervised or led by myself.

• Published:

1. *Crossover from adiabatic to sudden interaction quench in a Luttinger liquid.*
B. Dóra, M. Haque, and G. Zaránd, Phys. Rev. Lett. **106**, 156406 (2011).
2. *Analysis of a fully packed loop model arising in a magnetic Coulomb phase.*
L. D. C. Jaubert, M. Haque, R. Moessner, Phys. Rev. Lett. **107**, 177202 (2011).
3. *Three-vortex configurations in trapped Bose-Einstein condensates.*
J. A. Seman, E. A. L. Henn, M. Haque, R. F. Shiozaki, E. R. F. Ramos,
M. Caracanhas, P. Castilho, C. Castelo Branco, P. E. S. Tavares,
F. J. Poveda Cuevas, G. Roati, K. Magalhaes, and V. S. Bagnato;
Phys. Rev. A, **82**, 033616 (2010).
★ 10⁺ citations.
4. *Self-similar spectral structures and edge-locking hierarchy in open-boundary spin chains.* — M. Haque, Phys. Rev. A **82**, 012108 (2010).
5. *Disentangling Entanglement Spectra of Fractional Quantum Hall States on Torus Geometries.* — A. M. Laeuchli, E. J. Bergholtz, J. Suorsa, and M. Haque;
Phys. Rev. Lett. **104**, 156404 (2010).
★ 30⁺ citations.
6. *Entanglement Scaling of Fractional Quantum Hall states through Geometric Deformations.* — A. M. Laeuchli, E. J. Bergholtz, and M. Haque;
New Journal of Physics **12**, 075004 (2010).
★ 10⁺ citations.
7. *Interaction induced fractional Bloch and tunneling oscillations.*
R. Khomeriki, D. O. Krimer, M. Haque, and S. Flach;
Phys. Rev. A **81**, 065601 (2010).
8. *Finite-rate quenches of site bias in the Bose-Hubbard dimer.*
T. Venumadhav, M. Haque, and R. Moessner; Phys. Rev. B **81**, 054305 (2010).

9. *Entanglement between particle partitions in itinerant many-particle states.*
M. Haque, O. S. Zozulya, and K. Schoutens;
J. Phys. A: Math. Theor. **42**, 504012 (2009).
10. *Entanglement signatures of Quantum Hall phase transitions.* ▲
O. Zozulya, M. Haque, and N. Regnault; Phys. Rev. B **79**, 045409 (2009).
★ 15⁺ citations.
11. *Edge-localized states in quantum one-dimensional lattices.*
R. A. Pinto, M. Haque, and S. Flach; Phys. Rev. A **79**, 052118 (2009).
12. *Entanglement and level crossings in frustrated ferromagnetic rings.* ▲
M. Haque, J. N. Bandyopadhyay, and V. Ravi Chandra;
Phys. Rev. A **79**, 042317 (2009).
13. *Probing topological order in quantum Hall states using entanglement calculations.*
M. Haque; AMS Contemp. Math. **482**, 213 (2009).
14. *Survival benefits in mimicry: a quantitative framework.* ▲
A. Mikaberidze & M. Haque; J. Theor. Biol. **259**, 462 (2009).
15. *Particle partitioning entanglement in itinerant many-particle systems.*
O. Zozulya, M. Haque, and K. Schoutens; Phys. Rev. A **78**, 042326 (2008).
16. *A vortex dipole in a trapped two-dimensional Bose condensate.* ▲
W. Li, M. Haque and S. Komineas; Phys. Rev. A **77**, 053610 (2008).
★ 15⁺ citations.
17. *Symmetry-breaking Fermi surface deformations from central interactions in two dimensions.*
J. Quintanilla, M. Haque, and A. J. Schofield; Phys. Rev. B, **78**, 035131 (2008).
★ 10⁺ citations.
18. *Pomeranchuk instability: symmetry breaking and experimental signatures.*
J. Quintanilla, C. Hooley, B. J. Powell, A. J. Schofield, and M. Haque;
Physica B: Condensed Matter, **403**, 1279 (2008).
19. *Bipartite entanglement entropy in fractional quantum Hall states.*
O. Zozulya, M. Haque, K. Schoutens, and E. H. Rezayi;
Phys. Rev. B **76**, 125310 (2007).
★ 30⁺ citations.
20. *Entanglement entropy in fermionic Laughlin states.*
M. Haque, O. Zozulya, and K. Schoutens; Phys. Rev. Lett. **98**, 060401 (2007).
★ 50⁺ citations.
21. *Trapped fermionic clouds distorted from the trap shape due to many-body effects.*
M. Haque & H. T. C. Stoof; Phys. Rev. Lett., **98**, 260406 (2007).
★ 25⁺ citations.
22. *Deformation of a Trapped Fermi Gas with Unequal Spin Populations.*
G. B. Partridge, Wenhui Li, Y. A. Liao, R. G. Hulet, M. Haque, and H. T. C. Stoof;
Phys. Rev. Lett. **97**, 190407 (2006).
★ 160⁺ citations.

23. *Pairing of a trapped resonantly interacting fermion mixture with unequal spin populations.* — M. Haque & H. T. C. Stoof; Phys. Rev. A **74**, 011602 (2006).
★ 80⁺ citations.
24. *Ring-shaped luminescence pattern in biased quantum wells studied as a steady state reaction front.* — M. Haque; Phys. Rev. E **73**, 066207 (2006).
★ 10⁺ citations.
25. *Squeezing in the weakly interacting uniform Bose-Einstein condensate.*
M. Haque & A. E. Ruckenstein; Phys. Rev. A, **74**, 043622 (2006).
26. *Ultracold superstring in atomic boson-fermion mixtures.*
M. Snoek, M. Haque, S. Vandoren, and H. T. C. Stoof;
Phys. Rev. Lett. **95**, 250401 (2005).
★ 20 citations.
27. *Dynamics of a molecular Bose-Einstein condensate near a Feshbach resonance.*
M. Haque & H. T. C. Stoof; Phys. Rev. A **71**, 063603 (2005).
28. *Structural Transition of Wigner Crystal on Liquid Substrate.* ▲
M. Haque, I. Paul and S. Pankov; Phys. Rev. B **68**, 045427 (2003).
29. *Anomaly in the normalization constant of the (³He,p) reaction.*
M. Haque *et. al.*, Nuovo Cimento A, **111A**, 1131 (1998).
30. *⁶⁴Cu levels from the ⁶²Ni(³He,p) reaction at 18 MeV.*
A. K. Basak *et. al.*, Phys. Rev. C **56**, 1983 (1997).

• **Publicly available preprints (submitted or unpublished)**

31. *Strongly interacting one-dimensional bosons in arbitrary-strength optical lattices: from Bose-Hubbard to sine-Gordon and beyond.* ▲
A. Lazarides and M. Haque, arXiv:1112.3657.
32. *Slow interaction ramps in trapped many-particle systems: universal deviations from adiabaticity.* — M. Haque and F. E. Zimmer, arXiv:1110.0840. ▲
33. *Boundary-locality and perturbative structure of entanglement spectra in gapped systems.* — V. Alba, M. Haque, and A. M. Läuchli, arXiv:1107.1726.
34. *Non-adiabatic interaction ramps in a trapped Bose condensate.* ▲
F. E. Zimmer & M. Haque, arXiv:1012.4492.
35. *Transition Temperature of Dilute Weakly Repulsive Bose Gas.*
M. Haque & A. E. Ruckenstein, cond-mat/0212590.
36. *Weakly Non-ideal Bose Gas: Comments on Critical Temperature Calculations.*
M. Haque, cond-mat/0302076.